

Binary Information

Binary Name => Beginner Friendly

Language => C/C++

Arch => X86X64

Platform => Unix/Linux

```
$ file main
```

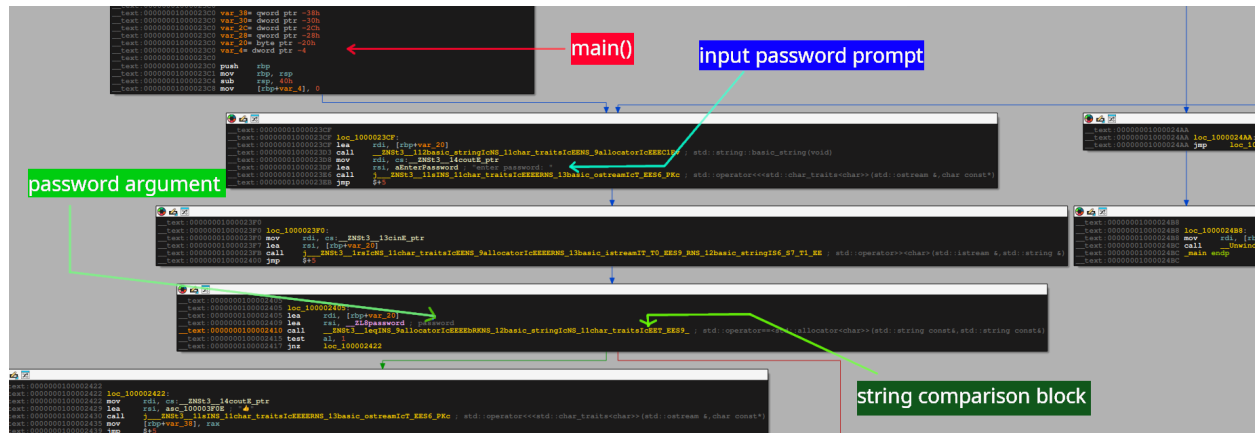
```
main: Mach-O 64-bit x86_64 executable, flags:<NOUNDEFS | DYLDLINK | TWOLEVEL |  
WEAK_DEFINES | BINDS_TO_WEAK | PIE>
```

Mach-O binaries basically run on iOS or Mac devices and since I am using Linux as reversing OS platform I cannot perform dynamic analysis. So static analysis will be performed and explained below how to understand the workflow and reverse engineer the binary.

Analysis

Static Analysis

Below is a provided image of the disassembly produced by IDA free 9.2 version in Linux distribution to a rough understanding of the program.



The above image gives us a rough idea about the working of the program.

Program Working

- Takes user string input
- Compares the user input with the predefined password string inside the program
- Enters the => “👍” block if the comparison is true

Of course better analysis could be performed but the binary format is not executable in Linux distributions.

Solving

The crackme challenge can be solved in 2 ways

- Using **IDA**
- Using **strings** command in Linux

IDA way

- Open IDA
- Go View tab
- View tab -> Open subviews -> Strings

Analyze the Strings windows carefully and try to find a unique string.

Address	Length	Type	String
HEADER:000...	00000010	C	__TEXT
HEADER:000...	00000010	C	__TEXT
HEADER:000...	00000010	C	__stubs
HEADER:000...	00000010	C	__TEXT
HEADER:000...	00000010	C	__init_offsets
HEADER:000...	00000010	C	__TEXT
HEADER:000...	00000010	C	__gcc_except_tab
HEADER:000...	00000010	C	__TEXT
HEADER:000...	00000010	C	__cstring
HEADER:000...	00000010	C	__TEXT
HEADER:000...	00000010	C	__unwind_info
HEADER:000...	00000010	C	__TEXT
HEADER:000...	00000010	C	__DATA_CONST
HEADER:000...	00000010	C	__got
HEADER:000...	00000010	C	__DATA_CONST
HEADER:000...	00000010	C	__DATA
HEADER:000...	00000010	C	__bss
HEADER:000...	00000010	C	__DATA
HEADER:000...	00000010	C	__LINKEDIT
HEADER:000...	0000000E	C	/usr/lib/dyld
HEADER:000...	00000018	C	/usr/lib/libc++.1.dylib
HEADER:000...	0000001B	C	/usr/lib/libSystem.B.dylib
__cstring:00...	00000009	C	buhnahna
__cstring:00...	00000011	C	enter password:

password!!!

So the password is **buhnahna**

strings command way

```
$ strings main | grep -i "buhnahna"
```

```
buhnahna
```